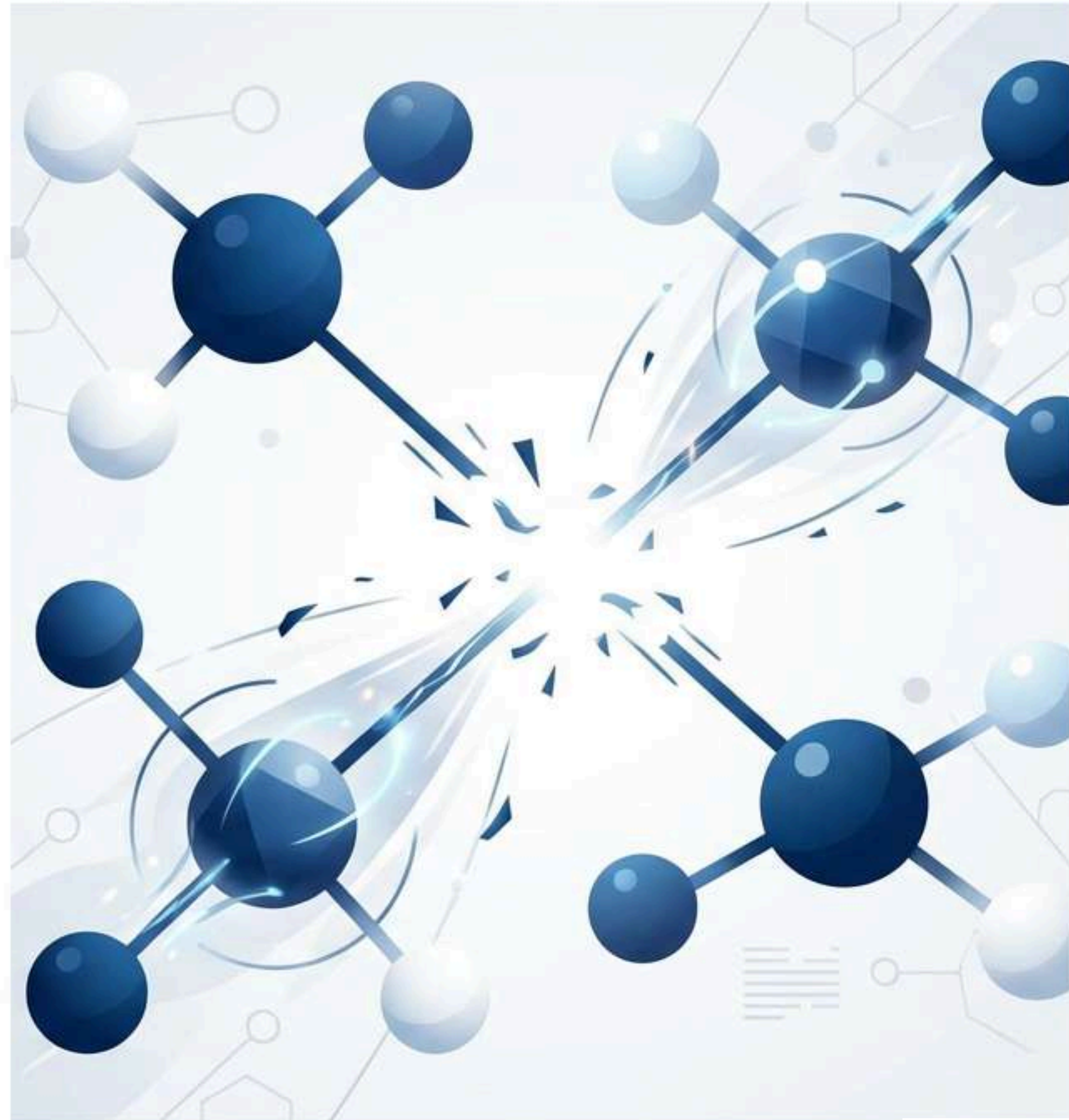


Chemical Reactions & Equations

The Language of Molecular Transformation

From the rusting of iron to the digestion of food, our world is defined by transformation. This deck decodes the syntax and grammar of these chemical changes.



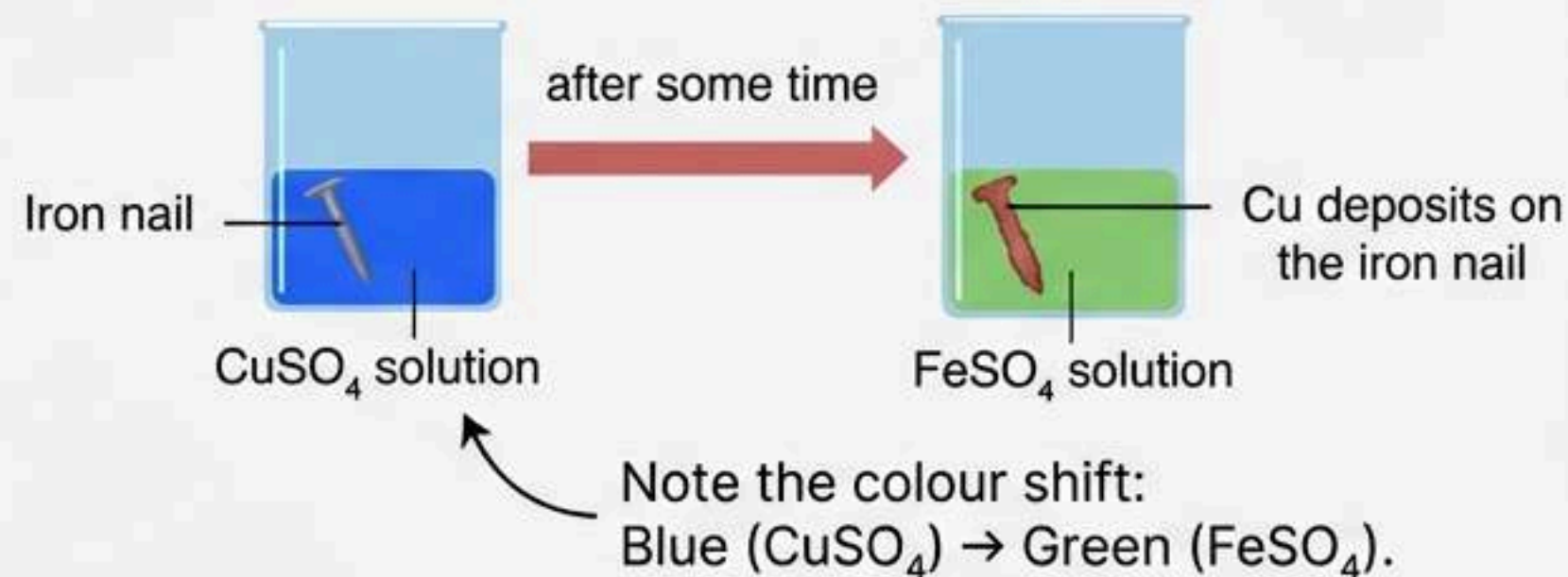
Physical Change



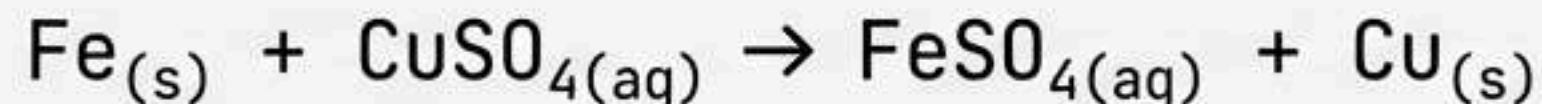
Change in colour or state, but no new substance is formed.
Example: Boiling water



Chemical Change



Reactants transform into products;
bonds break/make; new
substances formed.

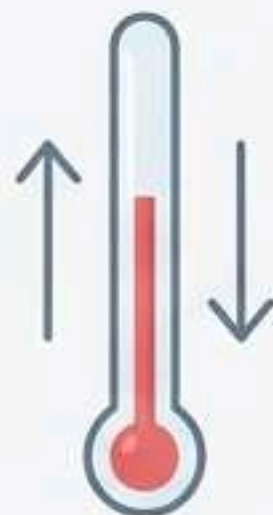


Observations that Determine a Chemical Reaction

How do we know a reaction has occurred?



Evolution of
gas



Change in
temperature



Formation of
precipitate



Change in
colour



Change of
state

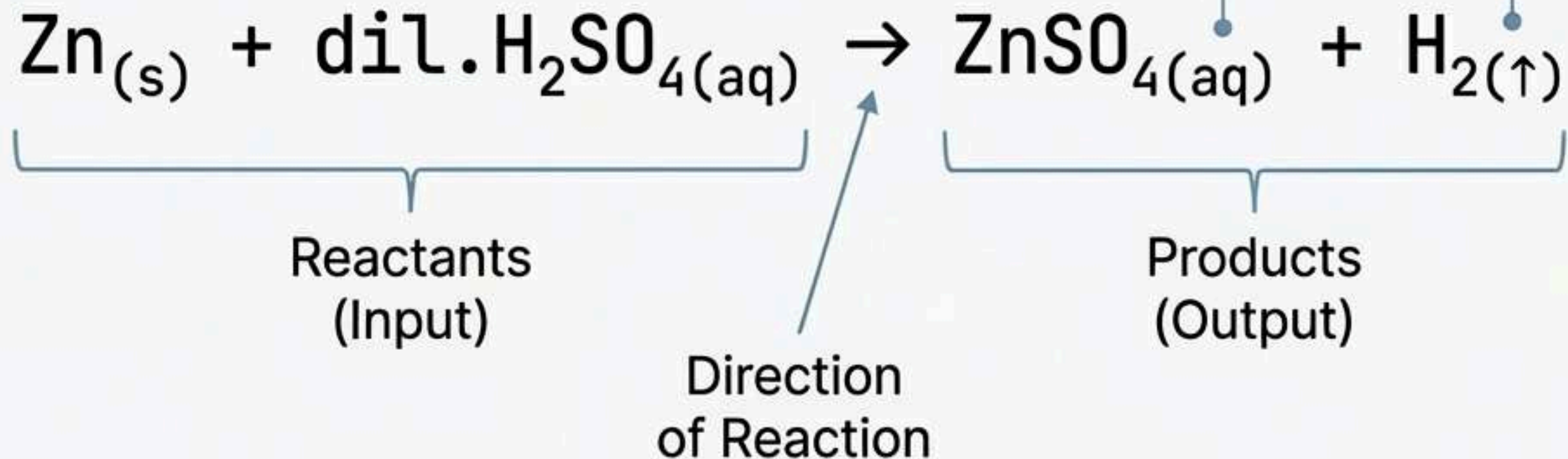
Chemical reactions are changes where reactants transform into products by making or breaking bonds between atoms.

Writing Chemical Equations

Word Equation:

Sodium + Chlorine → Sodium chloride

Chemical Equation: Uses symbols and formulae.



Legend

- (s) = Solid
- (l) = Liquid
- (aq) = Aqueous
(solution in water)
- (↑) = Gas
produced
- (↓) = Precipitate
formed

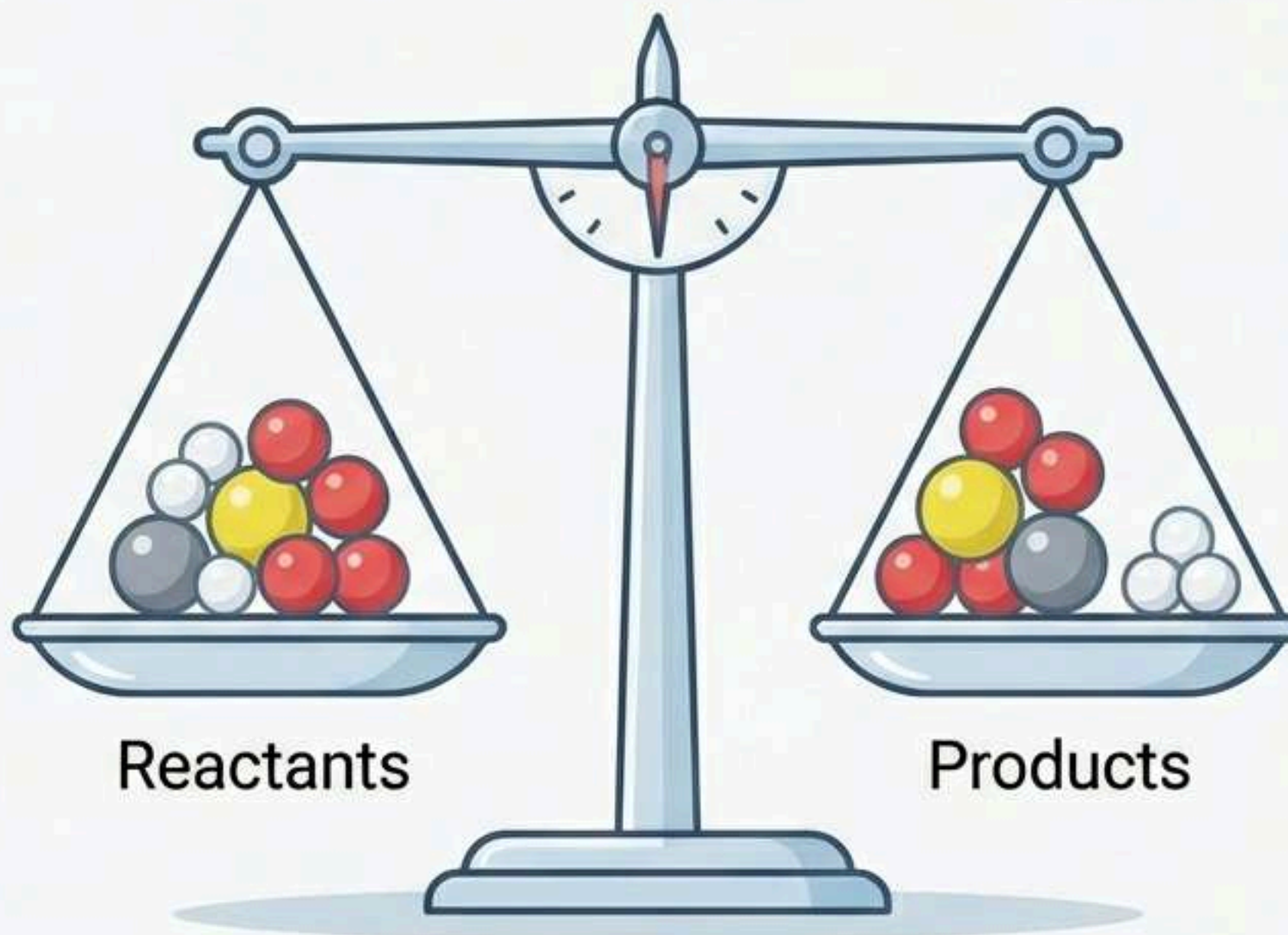
The Law of Conservation of Mass

The Law:

No atoms can be created or destroyed in a chemical reaction.

The Implication:

Total mass of products = Total mass of reactants.



Balanced Equation:

A chemical equation where the number of atoms of each element is equal on both sides.

Balancing: The Hit & Trial Method

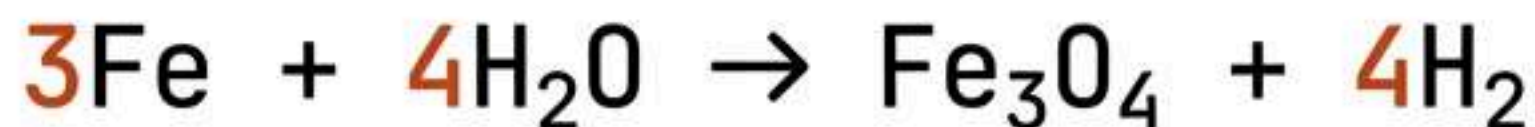
Step 1 (Unbalanced)



Count: Fe(1 vs 3), H(2 vs 2), O(1 vs 4). Unbalanced.



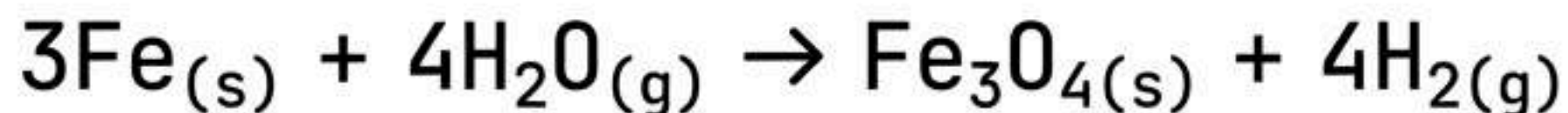
Step 2 (Adjusting)



Adjust coefficients to equalize atoms.



Step 3 (Balanced)



Final check: All atoms match.

Rule:

We change coefficients (numbers in front), never the subscripts (formula).

Masterclass: The Algebraic Balancing Technique

A systematic method for complex equations.

The Problem



Set Variables & Equations

Calcium (Ca): $a = 3c$

Carbon (C): $a = d$

Oxygen (O): $3a + 4b = 8c + 3d$

Hydrogen (H): $3b = 2d$

Phosphorus (P): $b = 2c$

Solve

Let $c = 1$

Then $a = 3(1) \rightarrow a = 3$

Since $a = d$, then $d = 3$

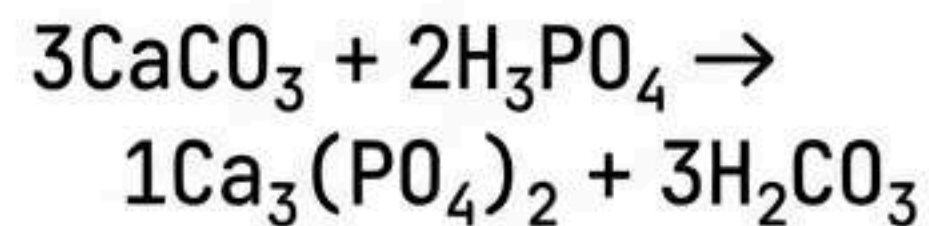
Since $b = 2c$, then $b = 2$



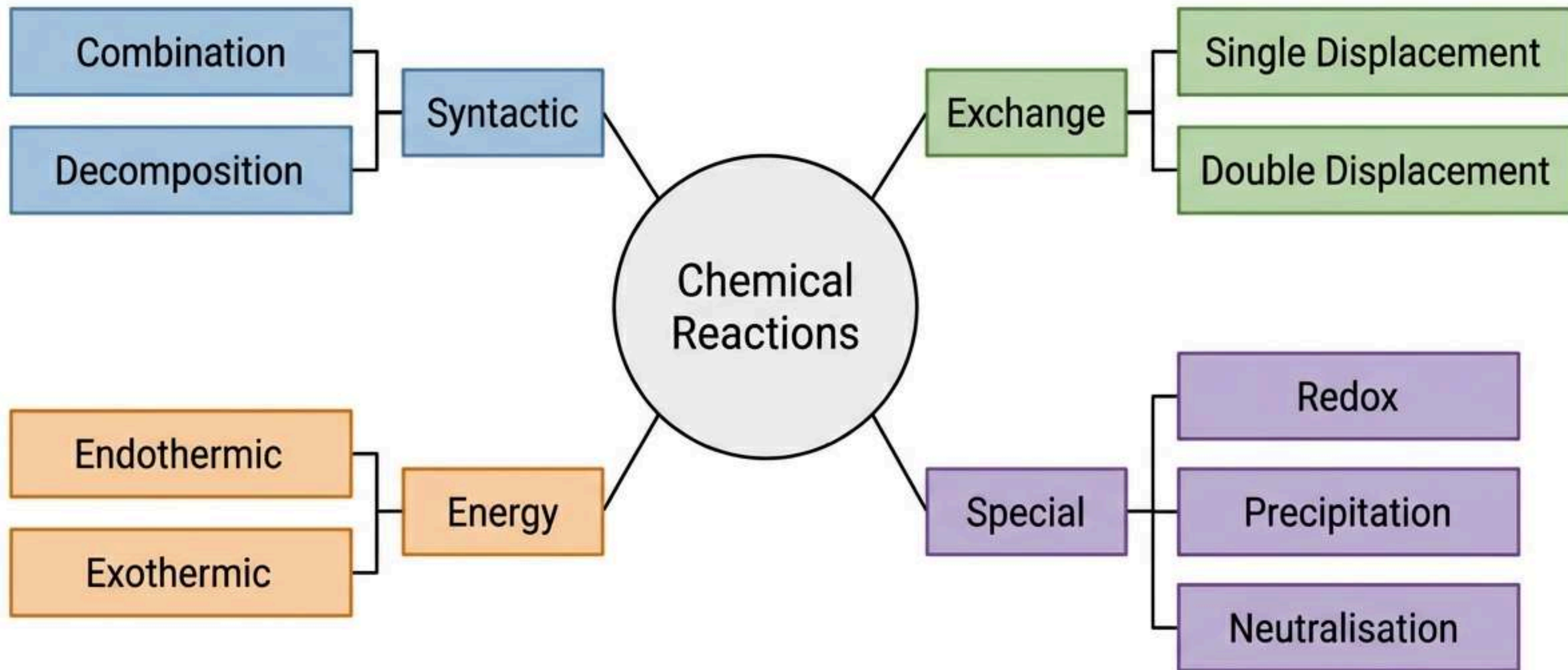
The Result



Substitute values back:



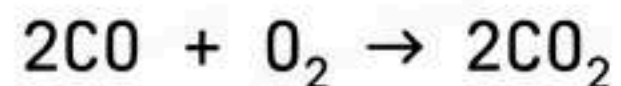
Types of Chemical Reactions



Synthesis and Analysis

Combination (Synthesis)

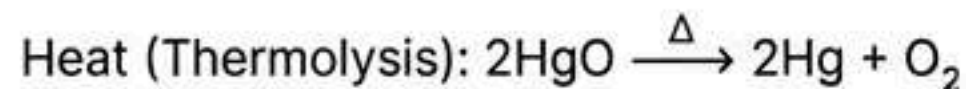
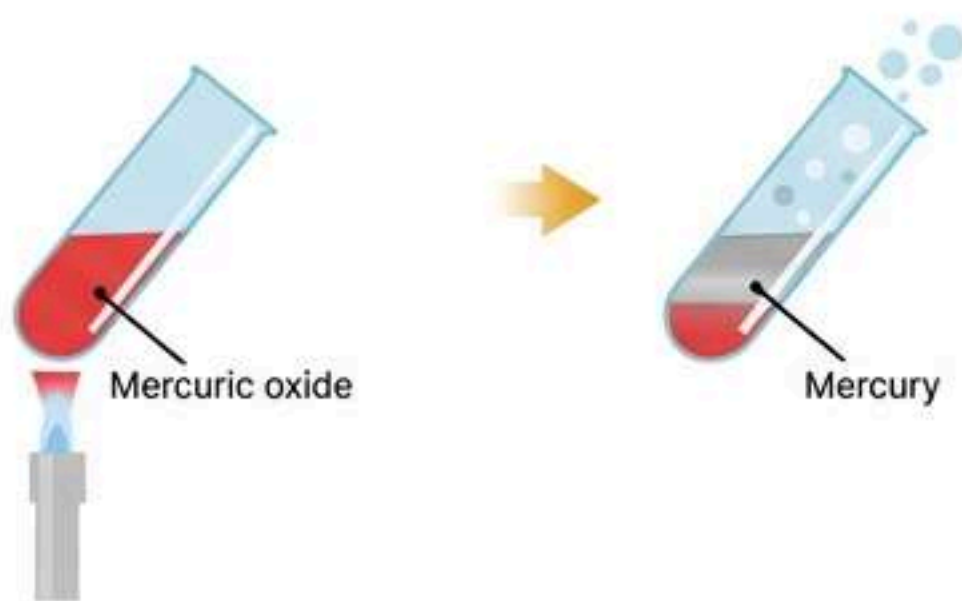
Two or more reactants form a single product ($A + B \rightarrow AB$).



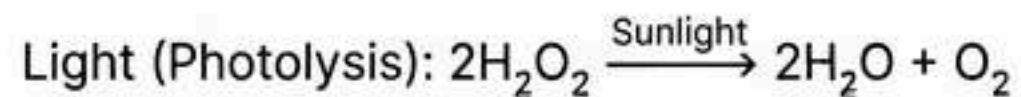
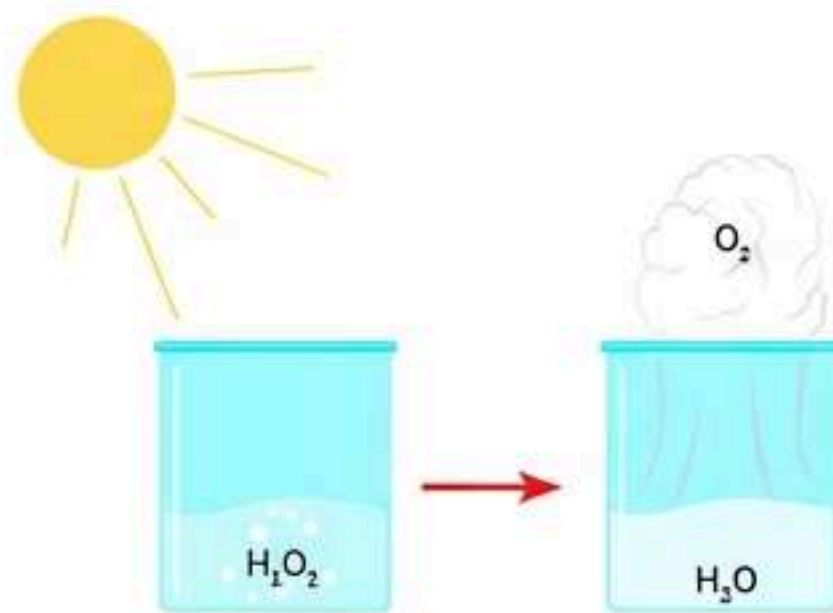
Decomposition (Analysis)

A single reactant breaks down into products ($AB \rightarrow A + B$). Three types:

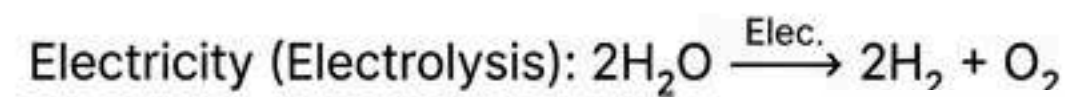
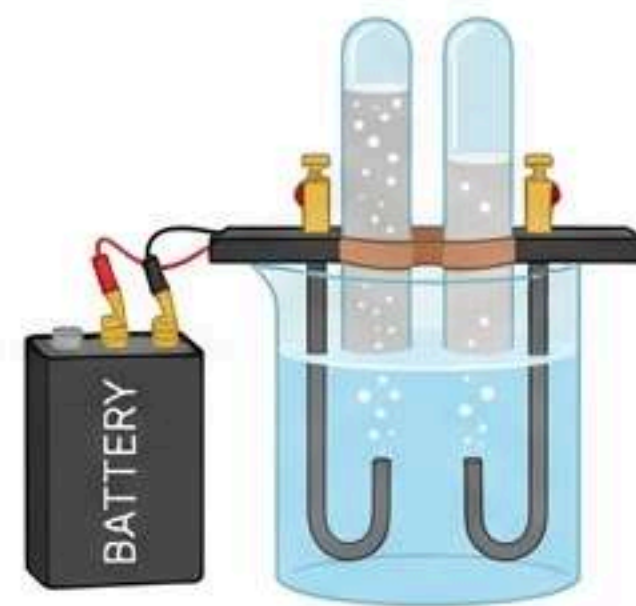
Thermal



Photo



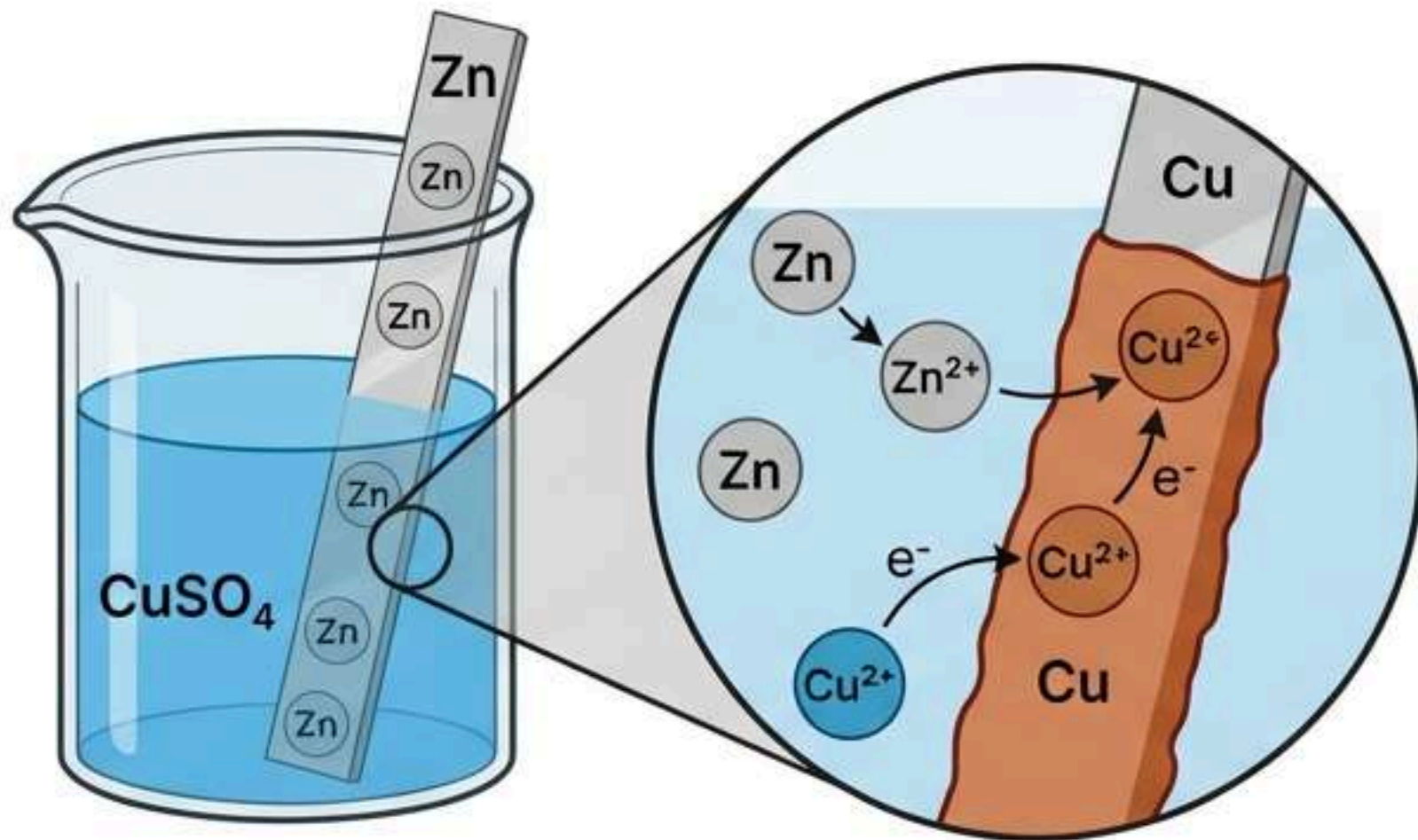
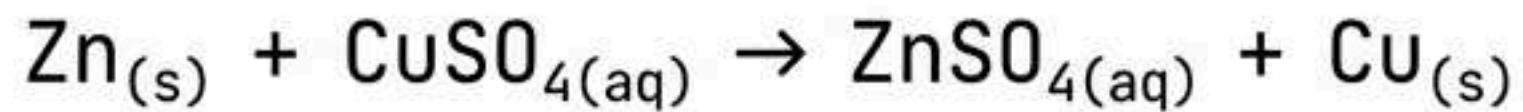
Electrolytic



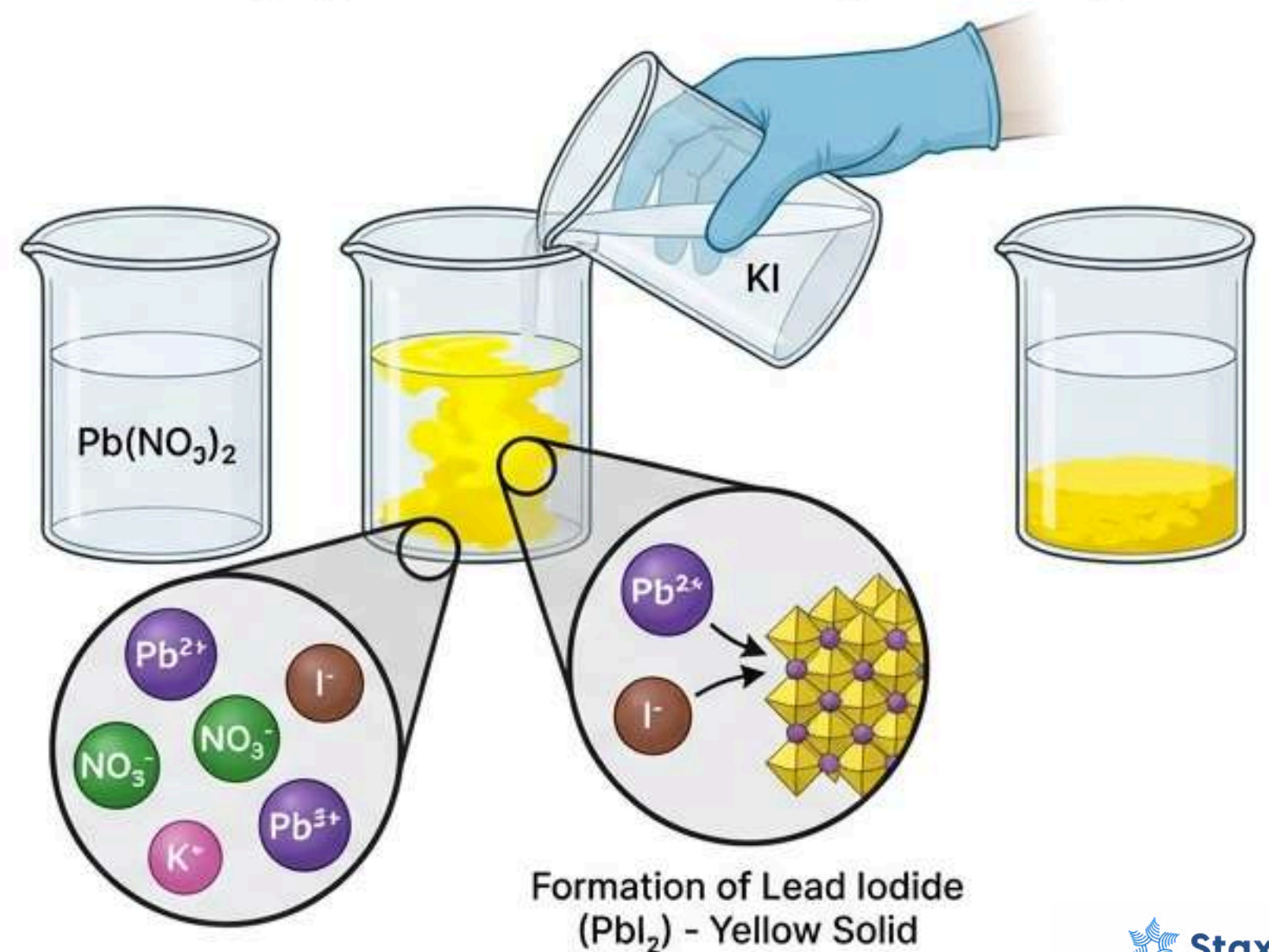
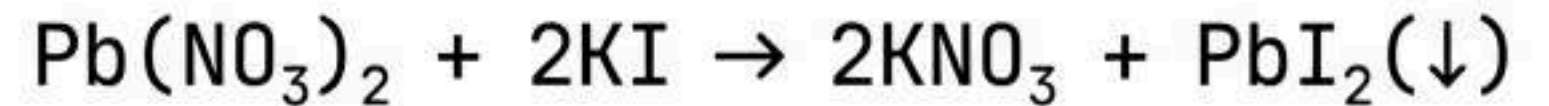
Displacement Reactions

The more reactive element takes the spot.

Single Displacement



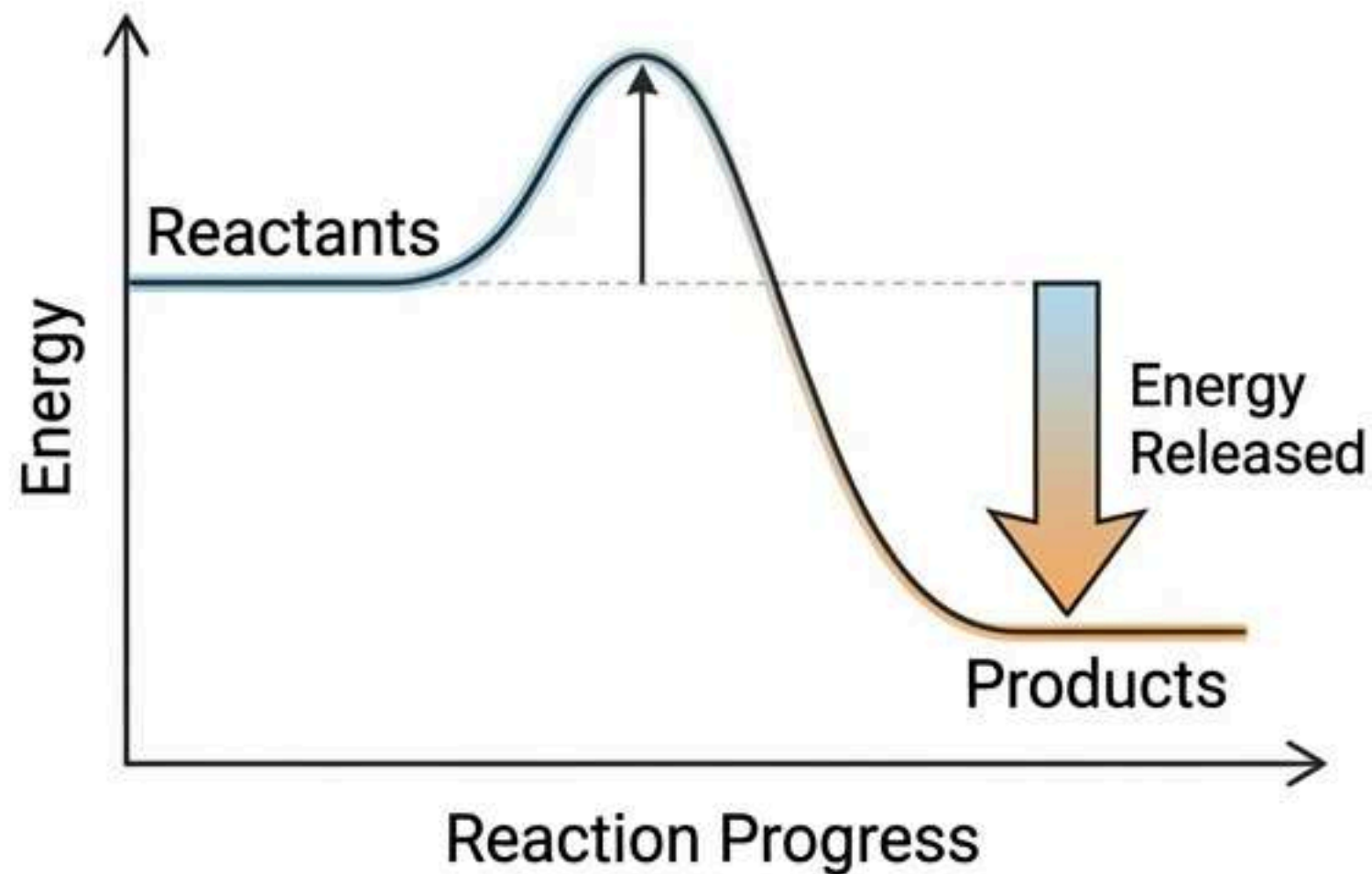
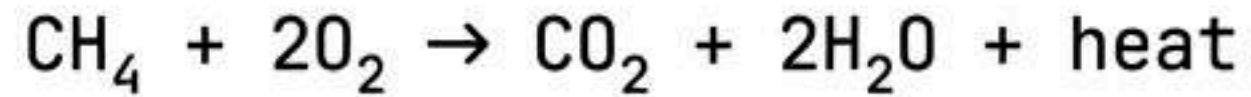
Double Displacement & Precipitation



The Energy Exchange

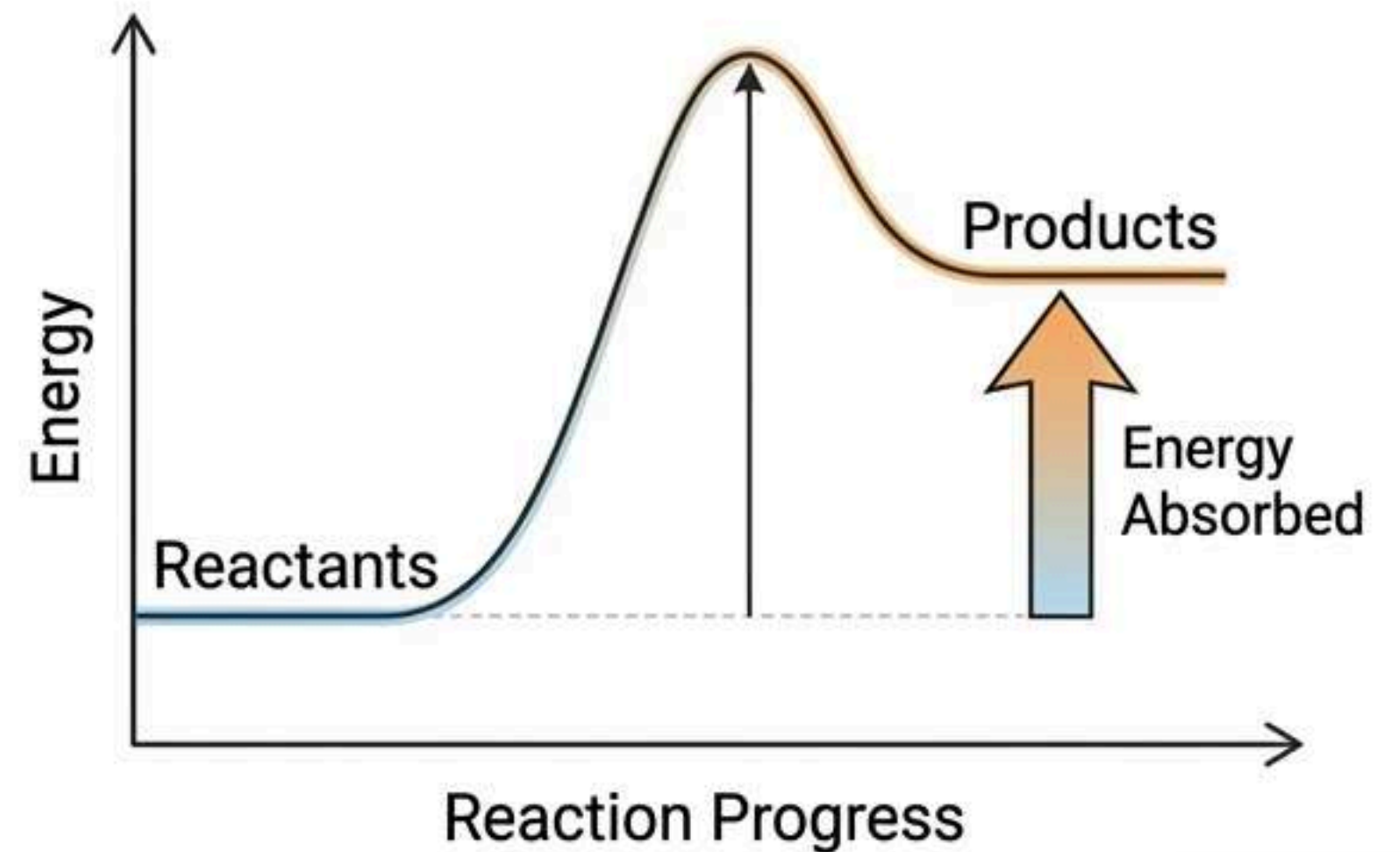
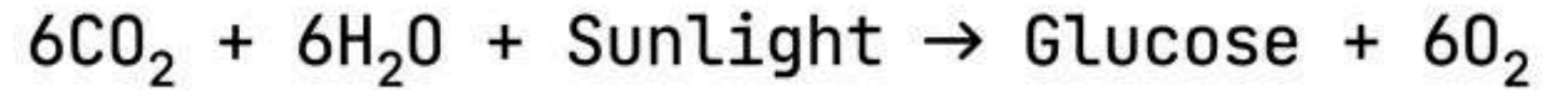
Exothermic Reaction

Heat is released (evolved).



Endothermic Reaction

Heat is absorbed (required).

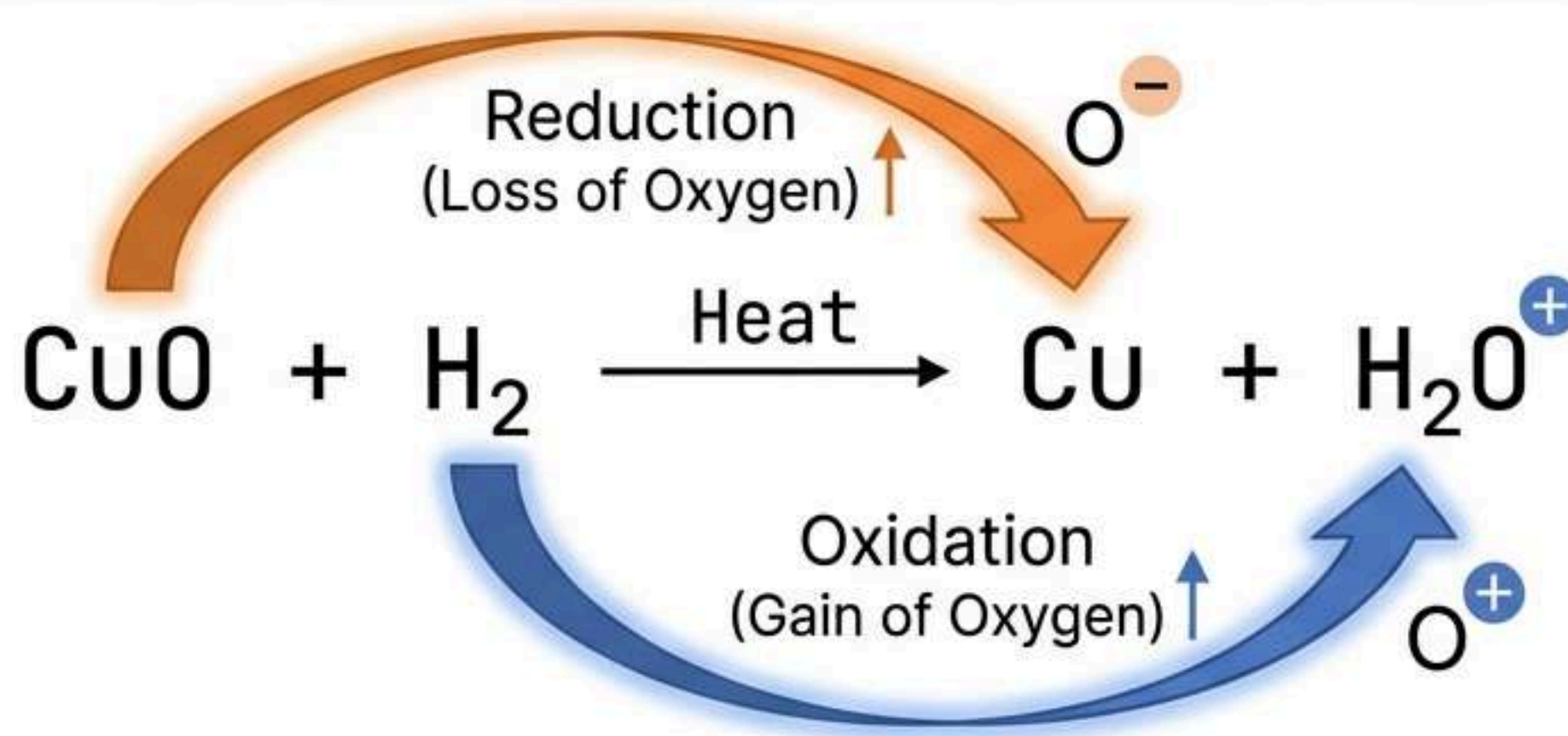


Redox Reactions

Reduction and Oxidation

Oxidation: Loss of Electrons | Gain of Oxygen | Loss of Hydrogen

Reduction: Gain of Electrons | Loss of Oxygen | Gain of Hydrogen



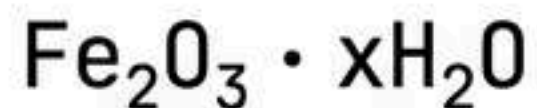
Oxidising Agent: CuO (Gets reduced)

Reducing Agent: H_2 (Gets oxidised)

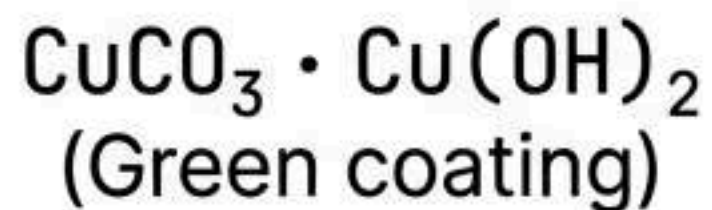
Real World Chemistry: Corrosion

Definition: "The gradual deterioration of metals by moisture, air, or chemicals."

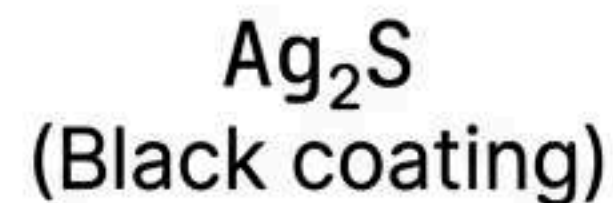
Rusting of Iron



Corrosion of Copper



Tarnishing of Silver



Real World Chemistry: Rancidity

Oxidation of fats and oils in food kept for long periods.

The Problem



Oxidation of fats and oils in food kept for long periods.

Impact: Foul smell, bad taste, potential infection.

The Solutions (Prevention)



Nitrogen Packaging: Flushing bags with inert gas.



Antioxidants: Additives that inhibit oxidation.



Physical Barriers: Air-tight containers & refrigeration.

Summary: The Cycle of Transformation

Change Chemical changes create new identities (New products). Physical changes do not.	The Law Mass is conserved. Atoms cannot be created or destroyed; equations must be balanced.
The Types Reactions vary in mechanism: Combination, Decomposition, Displacement, and Redox.	Impact Chemistry is not just theoretical—it drives energy systems, corrosion, and biological life.

Chemistry is the study of matter's ability to transform.